Computer Hardware & Maintenance

Ground to Mastery — Study Notes

Hardware, Software, Motherboards, Memory, Disks, Troubleshooting

CSC 121 | Year 1 Semester 2 | DCS

Coursework 40% (Test 10% + Assignment 10% + Practical 20%) | Final Exam 60%

How to Study This Course

This is a theory-heavy course with some practical components. Study by understanding components and how they relate to each other. Use your own computer as a reference — check your specs with real commands.

> ⚡ *In your terminal: lscpu (CPU info), free -h (memory), lsblk (disks), lspci (hardware), sudo dmidecode (full hardware info)*

> 📝 *Assessment: Test 10% + Assignment 10% + Practical 20% + Exam 60%. Know definitions, functions, and troubleshooting steps.*

UNIT 1: COMPUTER HARDWARE FUNDAMENTALS

Components, classification, system unit and peripherals

▸ 1.1 — What is a Computer?

A computer is an electronic device that accepts data (input), processes it according to instructions (programme), stores it, and produces output. It operates on the IPO cycle: Input → Process → Output.

Term / Concept	Definition / Explanation
`Hardware`	Physical components you can touch — CPU, RAM, HDD, keyboard, monitor
`Software`	Instructions that tell hardware what to do — OS, applications
`Firmware`	Software permanently stored in ROM — BIOS/UEFI — cannot be easily changed
`Data`	Raw facts and figures input to the computer
`Information`	Processed, meaningful data — the output

▸ 1.2 — Classification of Computers

Term / Concept	Definition / Explanation
`Supercomputer`	Most powerful — scientific research, weather forecasting, nuclear simulations
`Mainframe`	Large, powerful — banks, airlines for millions of transactions simultaneously
`Minicomputer (Server)`	Medium-sized — serves multiple users — web servers, file servers
`Microcomputer (PC)`	Personal computer — desktop, laptop, tablet — most common
`Embedded computer`	Inside other devices — microwave, car, ATM, washing machine

▸ 1.3 — System Unit Components

Term / Concept	Definition / Explanation
`CPU`	Central Processing Unit — the brain — fetches, decodes, executes instructions
`RAM`	Random Access Memory — temporary storage — data lost when power off — measured in GB
`ROM`	Read-Only Memory — permanent — stores BIOS/firmware
`Motherboard`	Main circuit board — connects all components
`PSU`	Power Supply Unit — converts AC mains to DC for components
`GPU`	Graphics Processing Unit — renders images — on motherboard or separate card
`NIC`	Network Interface Card — connects to network — wired (Ethernet) or wireless (WiFi)
`Sound Card`	Processes audio input/output
`Expansion slots`	PCIe, PCI slots on motherboard for adding GPU, NIC, sound cards

▸ 1.4 — Input Devices

- Keyboard — primary text input
- Mouse / touchpad — pointing and clicking
- Scanner — converts physical documents to digital
- Microphone — audio input
- Webcam — video input
- Barcode reader — reads barcodes
- Joystick / gamepad — gaming input
- Biometric readers — fingerprint, retina

▸ 1.5 — Output Devices

- Monitor (VDU) — visual output — LCD, LED, OLED
- Printer — produces paper output — inkjet, laser, dot-matrix, thermal
- Speakers — audio output
- Projector — large-screen display
- Plotter — precise drawing output for engineering

UNIT 2: COMPUTER SOFTWARE

Types of software, OS functions, installation

▸ 2.1 — Classification of Software

Term / Concept	Definition / Explanation

System Software	Manages hardware and provides platform for applications — OS, device drivers, utilities
Application Software	Does specific user tasks — Word, Excel, Firefox, Photoshop, games
Utility Software	Maintenance tools — antivirus, disk cleaner, file managers, backup tools
Device Drivers	Software that allows OS to communicate with specific hardware
Firmware	Software embedded in ROM — BIOS, printer firmware, router firmware

▸ 2.2 — Operating System Functions

1. Process management — runs and schedules programmes (multitasking)
2. Memory management — allocates RAM to running programmes
3. File system management — organises files in directories/folders
4. Device management — communicates with hardware through drivers
5. User interface — provides CLI (command line) or GUI (graphical)
6. Security — user accounts, permissions, encryption
7. Error handling — detects and reports system errors

▸ 2.3 — Types of Operating Systems

Term / Concept	Definition / Explanation
Windows	Proprietary GUI OS by Microsoft — Windows 11, Windows Server 2022
Linux/Ubuntu	Open-source — many distributions — your current OS — Ubuntu 24
macOS	Apple proprietary — based on Unix — only on Apple hardware
DOS	Disk Operating System — command-line only — early Microsoft OS
UNIX	Multi-user, multi-tasking — basis for Linux and macOS
Android / iOS	Mobile operating systems — Linux-based (Android), Unix-based (iOS)

▸ 2.4 — Software Licensing

Term / Concept	Definition / Explanation
Open Source	Source code freely available — can modify and redistribute — Linux, LibreOffice
Proprietary/Licensed	Cannot distribute or modify — must purchase licence — Windows, Adobe
Freeware	Free to use but source code closed — not open source

`Shareware`	Free trial — must pay after trial period
`GNU/GPL`	GNU General Public Licence — most common open source licence

UNIT 3: MOTHERBOARD / SYSTEM BOARD

Form factors, buses, BIOS, POST, CPU specs

▸ 3.1 — Motherboard Overview

The motherboard (system board) is the main printed circuit board that connects all components. Everything plugs into it directly or through cables.

▸ 3.2 — Form Factors

Term / Concept	Definition / Explanation
`ATX`	Standard full-size — most desktop PCs — 305mm x 244mm
`Micro-ATX`	Smaller version of ATX — fewer expansion slots
`Mini-ITX`	Very small — compact builds — limited expansion
`E-ATX`	Extended ATX — larger — high-end workstations/servers

▸ 3.3 — Computer Buses

Term / Concept	Definition / Explanation
`Data Bus`	Carries actual data between components — width = 32-bit or 64-bit
`Address Bus`	Carries memory addresses — determines addressable RAM
`Control Bus`	Carries control signals — read/write, interrupt signals
`Front Side Bus (FSB)`	Bus between CPU and motherboard chipset
`PCIe Bus`	PCI Express — modern high-speed bus — for GPU, NVMe SSD, NIC
`USB`	Universal Serial Bus — external device connection
`SATA`	Serial ATA — connects storage drives

▸ 3.4 — CPU Specifications

Term / Concept	Definition / Explanation
`Clock speed`	GHz — cycles per second — higher = faster (e.g. 3.8 GHz)
`Number of cores`	Multiple processors in one chip — 4-core, 8-core, 16-core
`Hyper-threading`	Intel technology — each core handles 2 threads — doubles

	logical CPUs
`Cache`	Very fast memory inside CPU — L1 (fastest, smallest) → L2 → L3 (slowest, largest)
`Turbo Boost`	Intel technology — automatically increases clock speed under load
`Socket type`	Physical connector between CPU and motherboard — must be compatible
`TDP`	Thermal Design Power — heat generated — determines cooling needed

▸ 3.5 — BIOS / UEFI and POST

Term / Concept	Definition / Explanation
`BIOS`	Basic Input/Output System — firmware in ROM — first code CPU runs on power-on
`UEFI`	Unified Extensible Firmware Interface — modern replacement for BIOS — faster boot
`CMOS`	Battery-backed memory — stores BIOS settings (date, time, boot order)
`POST`	Power-On Self Test — hardware check on startup — beep codes indicate errors
`Boot order`	BIOS setting — which device to boot from first (USB, HDD, CD)

> ⚡ *To enter BIOS: press Del, F2, or F12 (varies by manufacturer) during the first second of power-on.*

UNIT 4: PC MEMORY

RAM types, ROM types, virtual memory, memory management

▸ 4.1 — Memory Hierarchy (Fastest to Slowest)

Term / Concept	Definition / Explanation
`CPU Registers`	Fastest — inside CPU — few bytes — nanoseconds access
`L1/L2/L3 Cache`	Very fast — inside CPU — KB to MB — built-in SRAM
`RAM (Main Memory)`	Fast — separate chips — GB capacity — nanoseconds — data lost when off
`SSD Storage`	Slower — permanent — GB to TB — microseconds
`HDD Storage`	Slowest — permanent — GB to TB — milliseconds

▸ 4.2 — RAM Types

Term / Concept	Definition / Explanation

SRAM	Static RAM — fastest — used in CPU cache — expensive
DRAM	Dynamic RAM — main memory — needs refreshing — cheaper than SRAM
SDRAM	Synchronous DRAM — synced with CPU clock — older standard
DDR	Double Data Rate — transfers data on both clock edges — doubles speed
DDR4	4th generation DDR — current standard — 2133–3200+ MHz
DDR5	5th generation DDR — newest — 4800+ MHz — in newest systems
VRAM	Video RAM — on GPU — stores frame buffer and textures

▸ 4.3 — ROM Types

Term / Concept	Definition / Explanation
ROM	Read-Only Memory — factory-programmed — cannot be erased
PROM	Programmable ROM — written once by user — then permanent
EPROM	Erasable PROM — erased with UV light — reprogrammable
EEPROM	Electrically Erasable PROM — erased electrically — BIOS chips use this
Flash	Modern type of EEPROM — used in SSDs, USB drives, memory cards

▸ 4.4 — Virtual Memory & Swap

When RAM is full, the OS uses a portion of the hard disk as virtual memory (swap space). This is slower but prevents crashes. The file used is called a swap file or swap partition.

Memory commands in your terminal

```
# Check RAM and swap on Ubuntu
free -h

# Check swap details
swapon --show

# You added 4GB swap — your current swap total: ~7.6GB
# This prevents crashes when RAM fills up
```

UNIT 5: HARD DISK DRIVES

Storage technology, file systems, partitioning, disk management

▸ 5.1 — HDD Components

Term / Concept	Definition / Explanation
`Platters`	Magnetic disks that store data — spin at 5400-7200 RPM
`Read/write heads`	Reads and writes magnetic data on platters — one per platter surface
`Spindle motor`	Spins the platters at constant speed
`Actuator arm`	Moves read/write heads across platters to correct track
`Track`	Circular path on platter surface
`Sector`	Smallest unit of storage — typically 512 bytes or 4KB
`Cluster`	Group of sectors — minimum allocation unit in file system
`Cylinder`	All tracks at same position across all platters

▸ 5.2 — Storage Technologies

Term / Concept	Definition / Explanation
`HDD`	Hard Disk Drive — magnetic — mechanical — 5400/7200 RPM — cheap, large
`SSD`	Solid State Drive — flash memory — no moving parts — fast, silent, expensive
`NVMe SSD`	Non-Volatile Memory Express — SSD using PCIe slot — fastest storage
`IDE/PATA`	Older interface — parallel ATA — ribbon cable — obsolete
`SATA`	Serial ATA — current standard for HDDs and 2.5" SSDs — 6 Gb/s
`SCSI`	Small Computer Systems Interface — enterprise servers — very fast
`M.2`	Form factor for NVMe SSDs — plugs directly into motherboard slot

▸ 5.3 — File Systems

Term / Concept	Definition / Explanation
`FAT32`	File Allocation Table — Windows — max file 4GB — max partition 8TB
`NTFS`	New Technology File System — Windows — large files, permissions, encryption
`ext4`	Fourth extended filesystem — Linux — your Ubuntu uses ext4
`exFAT`	Extended FAT — cross-platform — used on USB drives and SD cards
`APFS`	Apple File System — macOS — snapshots, encryption

▸ 5.4 — Disk Maintenance

Term / Concept	Definition / Explanation
`Formatting`	Prepares disk for use — sets up file system — erases all data
`Partitioning`	Divides physical disk into logical sections — each can have own file system
`Defragmentation`	HDDs only — rearranges fragmented files — improves read speed
`Disk check`	Scans for errors in file system — chkdsk (Windows) / fsck (Linux)
`Disk caching`	OS stores recently read data in RAM for faster re-access
`RAID`	Redundant Array of Independent Disks — multiple disks for speed or redundancy

UNIT 6: COMPUTER TROUBLESHOOTING & MAINTENANCE

The 3 phases, common problems, tools, preventive maintenance

▸ 6.1 — The 3 Phases of Troubleshooting

1. Identify the problem — gather information, reproduce the fault, determine if hardware/software/user error
2. Fix the problem — research solution, apply fix, test thoroughly
3. Prevent recurrence — document what you did, implement preventive measures

▸ 6.2 — Common Hardware Problems & Solutions

Term / Concept	Definition / Explanation
`Computer won't power on`	Check power cable, check PSU switch, test PSU, check wall socket
`Screen blank / no display`	Check monitor cable and power, reseat GPU, test RAM, check POST beep codes
`HDD not detected`	Check SATA/power cables, check BIOS settings, HDD may have failed
`RAM failure`	Beeping on startup — reseat RAM modules, test one stick at a time
`Overheating / random shutdown`	Clean dust from fans and heatsinks, check thermal paste, improve airflow
`Computer very slow`	Check RAM usage (free -h), check for viruses, defragment HDD
`USB device not recognized`	Try different port, update drivers, check Device Manager
`Printer not printing`	Check cable/WiFi, check print queue, reinstall driver, check ink/toner

▸ 6.3 — Types of Maintenance

Term / Concept	Definition / Explanation
`Preventive`	Regular cleaning, updates, backups — done BEFORE problems occur
`Corrective`	Fixing problems AFTER they occur
`Predictive`	Monitoring for warning signs — SMART data, temperature, error logs
`Adaptive`	Updating system to work with new hardware/software changes

▸ 6.4 — Troubleshooting Tools

Term / Concept	Definition / Explanation
`Hardware tools`	Multimeter, screwdrivers, ESD strap, POST diagnostic card, cable testers
`Software tools (Windows)`	Device Manager, Task Manager, Event Viewer, chkdsk, sfc /scannow
`Software tools (Linux)`	dmesg, journalctl, lshw, lspci, lsusb, fsck, htop, smartctl
`Diagnostic software`	memtest86 (RAM), CrystalDiskInfo (HDD health), CPU-Z, GPU-Z

Hardware diagnostics — run these in your terminal

```
# Useful diagnostic commands on Ubuntu (your system)
lscpu                    # CPU information
free -h                  # RAM and swap usage
lsblk                    # block storage devices
lspci                    # PCI devices (GPU, NIC, etc.)
lsusb                    # USB devices
df -h                    # disk usage per partition
sudo dmesg | tail -20    # recent kernel messages/errors
sudo dmidecode -t memory # detailed RAM information
uptime                   # system uptime and load
sensors                  # CPU temperature (if installed)
```

UNIT 7: DOS COMMANDS

MS-DOS — internal and external commands, batch files

▸ 7.1 — Introduction to DOS

MS-DOS (Microsoft Disk Operating System) is a command-line operating system. It predates Windows. DOS commands are still used today in Windows Command Prompt and many concepts carry over to Linux/Bash.

▸ 7.2 — Internal DOS Commands (in COMMAND.COM)

Term / Concept	Definition / Explanation
`DIR`	List files and directories in current directory — like ls in Linux
`MD / MKDIR`	Make directory: MD foldername — creates a new folder
`CD / CHDIR`	Change directory: CD foldername CD.. CD\ CD C:\Users
`RD / RMDIR`	Remove directory — must be empty
`COPY`	Copy file: COPY source.txt destination.txt
`DEL`	Delete file: DEL filename.txt
`REN / RENAME`	Rename: REN oldname.txt newname.txt
`TYPE`	Display contents of text file: TYPE readme.txt
`CLS`	Clear screen — like clear in Linux
`DATE`	Display or set system date
`TIME`	Display or set system time
`VOL`	Display volume label of disk
`PATH`	Display or set executable search path
`VER`	Display DOS version

▸ 7.3 — External DOS Commands (separate .EXE files)

Term / Concept	Definition / Explanation
`FORMAT`	Format disk: FORMAT C: — DESTRUCTIVE — erases all data
`CHKDSK`	Check disk for errors: CHKDSK C: /F to fix errors
`FDISK`	Partition disk — creates, deletes, sets active partition
`DISKCOPY`	Copy entire floppy disk: DISKCOPY A: A:
`ATTRIB`	Set file attributes: ATTRIB +R file.txt — makes read-only
`LABEL`	Set disk volume label: LABEL C: SYSTEM
`SORT`	Sort output: DIR | SORT — pipe DIR output through SORT
`BACKUP`	Backup files to floppy
`EDIT`	Text editor for DOS
`MOVE`	Move file to different directory
`HELP`	Show help for any command: HELP DIR
`SYS`	Transfer system files to make bootable disk

▸ 7.4 — Batch File Programming

backup.bat — batch file example

```
@ECHO OFF
REM This is a comment — REM = REMark
ECHO Starting backup...

REM Variables
SET SOURCE=C:\Documents
SET DEST=D:\Backup

REM Create destination if not exists
IF NOT EXIST %DEST% MD %DEST%

REM Copy files
COPY %SOURCE%\*.* %DEST%

REM Conditional
IF EXIST output.txt ECHO File exists!

REM Loop
FOR %%i IN (*.txt) DO ECHO Found: %%i

ECHO Backup complete!
PAUSE
```

> 📝 *Exam: Know the difference between internal and external DOS commands. Internal commands are in COMMAND.COM. External commands are separate .EXE or .COM files on disk.*

UNIT 8: COMPUTER SERVER HARDWARE & NETWORKS

Server types, server components, client-server model

▸ 8.1 — Types of Servers

Term / Concept	Definition / Explanation
`Tower server`	Looks like a desktop tower — small organisations — easy to maintain
`Rack server`	Mounted in standard rack unit — data centres — 1U, 2U, 4U height
`Blade server`	Very thin — multiple blades in chassis — high density — enterprise
`Mainframe`	Largest — processes millions of transactions — banks, airlines

▸ 8.2 — Client-Server Model

A client requests a service. A server provides the service. Multiple clients can connect to one server simultaneously.

Term / Concept	Definition / Explanation

`Web server`	Hosts websites — Apache, Nginx — serves HTTP requests
`File server`	Stores and shares files — NAS (Network Attached Storage)
`Mail server`	Handles email — sends (SMTP) and receives (POP3/IMAP) email
`Database server`	Stores databases — MySQL, PostgreSQL, MSSQL
`DNS server`	Translates domain names to IP addresses
`DHCP server`	Assigns IP addresses automatically to network devices

▸ 8.3 — SCSI (Small Computer Systems Interface)

SCSI is a high-speed interface for connecting storage devices and other peripherals. Used in servers because it supports more devices and is faster than IDE/SATA. SCSI devices are daisy-chained and each needs a unique ID (0-7 or 0-15).